

HeteroFormer: Tightly Coupling Sequence Prototypes with Heterogeneous Feature Interaction for CTR Prediction

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Abstract

Click-through rate (CTR) prediction demands jointly modeling long-range user behavior sequences and heterogeneous non-sequential features under a unified representation. Existing architectures predominantly adopt a decoupled pipeline: sequences are first compressed into fixed-length vectors, which are then fused with dense features through token-mixing modules. This separation limits both the depth of sequential structure utilization and the flexibility of cross-field interaction. We present **HeteroFormer**, a unified architecture that tightly couples continuous sequence modeling with prototype-conditioned heterogeneous feature interaction. The core insight is to represent sequential behavior as a *dynamic prototype manifold*—a set of learnable semantic anchors whose geometry is conditioned on user features via Cayley rotation and assigned through entropy-regularized optimal transport (Sinkhorn). These prototypes then serve as soft biases and gating signals within a cross-field interaction network, enabling sequence-derived semantics to directly shape feature interactions rather than being injected as passive context. Furthermore, we introduce a *generative semantic layer* that captures residual information via an information-bottleneck diffusion explainer and an energy-based error predictor; this layer is trained through *decoupled semantic optimization*, where generative objectives are isolated from the primary CTR path via gradient detachment and governed by an overfitting-aware meta-controller. We further provide an in-depth analysis on why the generative pathway naturally induces *semantic IDs* that render the model interpretable, why traditional Transformer-based interaction is insufficient for heterogeneous CTR features, and how prototype-driven assignment redefines explainability in modern recommendation systems. Extensive theoretical analysis and architectural ablations demonstrate that HeteroFormer achieves richer interaction capacity while maintaining favorable computational complexity. **Code is available at:** https://github.com/zhuizhuzheming/taac_2026.

1 Introduction

Industrial large-scale recommendation models (LRMs) face the fundamental challenge of jointly modeling long-range user behavior sequences and heterogeneous non-sequential features under strict representational constraints. The prevailing paradigm follows a two-stage

pipeline: (1) a sequence compressor such as DIN [1], BST [3], or LONGER [4] first aggregates variable-length behavior sequences into fixed-length query tokens; (2) a feature interaction module such as FM [6], DeepFM [7], or RankMixer [9] subsequently mixes these tokens with non-sequential features. While modular, this decoupled design inherently creates a representational bottleneck: sequential information is compressed once and for all before interaction, and cross-field interactions cannot feed back to refine sequence representations.

Recent works have attempted to bridge this gap. HyFormer [5] proposes an alternating optimization between query decoding and query boosting, yet still treats sequence and interaction as distinct phases. UniMixer [11] unifies attention-based, TokenMixer-based, and factorization-machine-based scaling blocks under a parameterized token-mixing framework, but does not explicitly address how sequential structure should inform the mixing pattern. TokenMixer-Large [10] scales up mixing-and-reverting paradigms with sparse per-token MoE, while MixFormer [12] and TokenFormer [13] explore unified Transformer-style architectures that jointly model sequential behaviors and multi-field features. Despite these advances, a principled mechanism that allows *sequence-derived semantics to directly parameterize heterogeneous feature interactions* remains under-explored.

We identify three key limitations of the current paradigm:

1. **Passive sequence injection.** Compressed sequence vectors are typically concatenated or added to non-sequential tokens, acting as “dead” context rather than actively shaping interaction weights.
2. **Rigid feature interaction.** Token-mixing modules apply fixed permutation patterns or dense attention, lacking conditional adaptation based on user-specific sequential patterns.
3. **Under-utilized sequential structure.** High-level semantic concepts latent in behavior sequences (e.g., category-level preferences, temporal intent shifts) are not explicitly disentangled and leveraged as interpretable control signals.

To address these limitations, we propose **HeteroFormer**, whose contributions are summarized as follows:

- We introduce a **Dynamic Prototype Manifold** that represents behavior sequences as soft assignments over learnable semantic prototypes. The prototype

geometry is conditioned on user features through Cayley rotation, and assignment is formulated as entropy-regularized optimal transport via Langevin-Sinkhorn, providing differentiable, sparse, and interpretable sequence summarization.

- We propose **Proto-Conditioned Cross-Field Interaction**, where prototype assignments serve as soft biases in multi-head cross-field attention and as gating signals in FiLM-conditioned feed-forward networks. This tightly couples sequential semantics with heterogeneous feature interaction in a bidirectional, co-evolutionary manner.
- We design a **Generative Semantic Layer** comprising a Diffusion Explainer (information-bottleneck residual encoder) and an Energy Calibrator (error-predictive scoring function). These modules capture complementary residual signals without interfering with the primary CTR gradient path. Importantly, we demonstrate that this generative pathway naturally induces **semantic IDs**—discrete, human-interpretable indices over the prototype manifold—that render the model’s sequential reasoning transparent.
- We present **Decoupled Semantic Optimization**, a training methodology that isolates generative and discriminative objectives through gradient detachment and governs their coupling via an overfitting-aware MetaAligner, ensuring stable co-training.
- We provide a critical analysis on why **standard Transformer-based interaction is ill-suited** for heterogeneous CTR features, and argue that prototype-driven assignment redefines explainability in modern recommendation systems by replacing opaque attention weights with semantically grounded mixture coefficients.

2 Related Work

Sequence Modeling in CTR Prediction. Early approaches such as DIN [1] and DIEN [2] employ attention mechanisms to activate relevant behaviors with respect to target items. Transformer-based methods including BST [3] and LONGER [4] extend this to multi-layer self-attention over long sequences. State-space models (SSMs) have recently gained traction for their linear complexity; structured variants such as S4 [14] and Mamba [15] offer principled long-range dependency modeling. HeteroFormer adopts a lightweight SSM cell for continuous-time sequence encoding, coupled with multi-scale aggregation to capture both short-term and long-term user intent.

Heterogeneous Feature Interaction. Factorization machines [6] and their deep variants (DeepFM [7], xDeepFM [8]) model second- and high-order feature

interactions explicitly. TokenMixer [9] and its large-scale extension TokenMixer-Large [10] reformulate interaction as token-mixing via learned permutation matrices. UniMixer [11] unifies attention, TokenMixer, and FM under a generalized parameterized mixing framework with Sinkhorn-Knopp doubly-stochastic constraints. HyFormer [5] alternates between query decoding (sequence attention) and query boosting (token mixing). HeteroFormer differs by making the *interaction pattern itself* conditional on sequence-derived prototypes, rather than applying a fixed mixing operator.

Generative and Energy-Based Models in Recommendation. Diffusion models have been explored for sequential recommendation [16] and counterfactual explanation generation. Energy-based models (EBMs) provide a natural framework for uncertainty quantification and out-of-distribution detection. HeteroFormer draws inspiration from these directions but employs them in a *residual, auxiliary* capacity: the Diffusion Explainer compresses residual sequence information through an information bottleneck, while the Energy Calibrator predicts prediction error magnitude to enable dynamic logit calibration at inference time.

3 Methodology

3.1 Overview

The HeteroFormer architecture consists of four tightly integrated stages (Figure 1):

1. **Multi-View Encoding:** User, item, and dense features are encoded into shared semantic representations with task-specific projections.
2. **Continuous Sequence Modeling:** Variable-length behavior sequences are processed by SSM cells with continuous-time encoding, producing multi-scale summaries (short-term, long-term, static) that are fused across domains.
3. **Dynamic Prototype Manifold:** The fused sequence representation is mapped onto a user-conditioned prototype manifold via Cayley rotation and Langevin-Sinkhorn optimal transport, yielding sparse prototype assignments and a compact prototype representation.
4. **Proto-Conditioned Interaction & Prediction:** Prototype assignments bias cross-field attention and gate feed-forward transformations within the interaction network; a calibrated prediction head fuses static, prototype, and residual signals for the final CTR estimate.

3.2 Multi-View Feature Encoding

Let $\mathbf{x}_u \in \mathbb{R}^{d_u}$, $\mathbf{x}_i \in \mathbb{R}^{d_i}$, $\mathbf{x}_u^{\text{dense}} \in \mathbb{R}^{d_u^{\text{den}}}$, and $\mathbf{x}_i^{\text{dense}} \in \mathbb{R}^{d_i^{\text{den}}}$ denote user ID features, item ID features, user

dense features, and item dense features, respectively. Each ID feature j with vocabulary size V_j is embedded via a constrained embedding layer with max-norm projection:

$$\begin{aligned} \mathbf{e}_j &= \text{Embed}_{\text{max-norm}}(x_j; V_j, d_e) \\ &= \frac{\mathbf{W}_j \mathbf{o}(x_j)}{\|\mathbf{W}_j \mathbf{o}(x_j)\|_2} \cdot \gamma, \end{aligned} \quad (1)$$

where $\mathbf{o}(x_j)$ is the one-hot indicator, $\mathbf{W}_j \in \mathbb{R}^{d_e \times V_j}$, and γ is the maximum norm. For multi-slot features, we average slot embeddings. Dense features are linearly projected to dimension d_m .

User and item tokens are aggregated into field-wise representations $\mathbf{z}_u, \mathbf{z}_i \in \mathbb{R}^{d_m}$. A shared context vector is formed as $\mathbf{z}_{\text{ctx}} = \mathbf{z}_u + \mathbf{z}_i$, and task-specific views are generated via shallow MLPs for CTR, diffusion, and energy tasks. Formally:

$$\begin{aligned} \mathbf{z}_{\text{shared}} &= \text{LayerNorm}(\mathbf{z}_u + \mathbf{z}_i + \mathbf{z}_{\text{ctx}}), \\ \mathbf{z}_{\text{view}}^{(t)} &= \text{MLP}_{\text{view}}^{(t)}(\mathbf{z}_{\text{shared}}), \end{aligned} \quad (2)$$

where $t \in \{\text{ctr}, \text{diff}, \text{energy}\}$.

3.3 Continuous Sequence Modeling

Behavior sequences are heterogeneous in both feature type and temporal granularity. For each sequence domain s , we embed discrete feature IDs into a concatenated representation $\mathbf{h}_{s,\ell}^{(0)} \in \mathbb{R}^{d_m}$ at position ℓ .

3.3.1 SSM Cell with Continuous-Time Discretization

We employ a structured state-space cell inspired by S4 [14]. Given input $\mathbf{x}_\ell \in \mathbb{R}^{d_m}$ and continuous time-step $\Delta_\ell > 0$, the state $\mathbf{h}_\ell \in \mathbb{R}^{d_s}$ evolves as:

$$\begin{aligned} \bar{\mathbf{A}}_\ell &= \exp(\Delta_\ell \cdot \mathbf{A}), \\ \bar{\mathbf{B}}_\ell &= \Delta_\ell \cdot \mathbf{B}, \\ \mathbf{h}_\ell &= \bar{\mathbf{A}}_\ell \odot \mathbf{h}_{\ell-1} + \bar{\mathbf{B}}_\ell \mathbf{x}_\ell, \\ \mathbf{y}_\ell &= \mathbf{C} \mathbf{h}_\ell + D \mathbf{x}_\ell, \end{aligned} \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{d_s \times d_s}$ is diagonal with negative real entries initialized as $\mathbf{A}_{kk} = -\exp(k)$, $\mathbf{B} \in \mathbb{R}^{d_s \times d_m}$, $\mathbf{C} \in \mathbb{R}^{d_m \times d_s}$, and D is a learned skip scalar. In practice, we parameterize Δ_ℓ via a softplus projection of inter-event time differences:

$$\Delta_\ell = \text{softplus}(\text{MLP}(\tau_\ell - \tau_{\ell-1})). \quad (4)$$

3.3.2 Multi-Scale Aggregation and Cross-Domain Fusion

After L SSM layers, we derive three complementary summaries:

- **Short-term:** Exponentially weighted pooling over recent positions,

$$\mathbf{s}_{\text{short}} = \sum_{\ell} w_\ell^{(s)} \mathbf{y}_\ell, \quad (5)$$

with $w_\ell^{(s)} \propto \exp(-\ell/\tau_s) \cdot \mathbb{I}[\ell < \ell_{\text{max}}]$.

- **Long-term:** Uniform mean pooling over valid positions,

$$\mathbf{s}_{\text{long}} = \frac{1}{\ell_{\text{max}}} \sum_{\ell=1}^{\ell_{\text{max}}} \mathbf{y}_\ell. \quad (6)$$

- **Static:** A learned projection of mean, variance, and last-state statistics:

$$\mathbf{s}_{\text{static}} = \text{MLP}_{\text{stat}}([\bar{\mathbf{y}}; \boldsymbol{\sigma}_{\mathbf{y}}; \mathbf{y}_{\ell_{\text{max}}}]). \quad (7)$$

When multiple sequence domains exist (e.g., click, purchase, search), domain-specific summaries are projected and fused via cross-domain attention with a learnable query token:

$$\mathbf{z}_{\text{seq}} = \text{MultiHeadAttn}(\mathbf{q}_{\text{fuse}}, [\mathbf{s}_{\text{short}}^{(1)}, \dots, \mathbf{s}_{\text{short}}^{(S)}]), \quad (8)$$

where $\mathbf{q}_{\text{fuse}} \in \mathbb{R}^{d_m}$ is a learned global query.

3.4 Dynamic Prototype Manifold

The sequence representation $\mathbf{z}_{\text{seq}}$ is typically high-dimensional and noisy. We propose to map it onto a *prototype manifold*—a set of K learnable semantic anchors $\{\boldsymbol{\eta}_k\}_{k=1}^K \subset \mathbb{S}^{d_m-1}$ (unit hypersphere). The manifold geometry is conditioned on the user feature $\mathbf{z}_u$ through a Cayley rotation, and assignment is computed via entropy-regularized optimal transport.

3.4.1 User-Conditioned Cayley Rotation

To adapt prototypes to individual users while preserving the manifold structure, we apply a low-rank skew-symmetric update. Let $\mathbf{U}, \mathbf{V} \in \mathbb{R}^{d_m \times r}$ be learnable rank- r factors, and let $\mathbf{w} = \tanh(\text{MLP}(\mathbf{z}_u)) \in \mathbb{R}^r$ be user-specific scaling coefficients. The skew-symmetric matrix is:

$$\mathbf{G}(\mathbf{z}_u) = \mathbf{U} \text{diag}(\mathbf{w}) \mathbf{V}^\top - \mathbf{V} \text{diag}(\mathbf{w}) \mathbf{U}^\top. \quad (9)$$

The Cayley transform yields an orthogonal rotation matrix:

$$\mathbf{R}(\mathbf{z}_u) = (\mathbf{I} + \frac{1}{2} \mathbf{G})(\mathbf{I} - \frac{1}{2} \mathbf{G})^{-1}, \quad (10)$$

and the user-conditioned prototypes are:

$$\boldsymbol{\mu}_k(\mathbf{z}_u) = \mathbf{R}(\mathbf{z}_u) \boldsymbol{\eta}_k, \quad \boldsymbol{\mu}_k \in \mathbb{S}^{d_m-1}. \quad (11)$$

This ensures that prototype norms are preserved and the transformation is invertible, avoiding representational collapse.

3.4.2 Concentration-Modulated Similarity

The affinity between the sequence representation and each prototype is modulated by a user- and time-dependent concentration parameter κ . We parameterize:

$$\begin{aligned} \kappa(\mathbf{z}_u, \Delta t) &= \text{softplus}(\kappa_{\text{base}} + \text{MLP}_{\kappa}(\mathbf{z}_u)) \\ &\quad \cdot \exp(-\lambda_{\kappa} \Delta t / 86400) + \kappa_{\text{min}}, \end{aligned} \quad (12)$$

where Δt is the time span of the sequence (in seconds), and λ_κ is a learned decay rate. This captures the intuition that older sequences should yield more diffuse prototype assignments.

The unnormalized log-probability (logit) of assignment is:

$$\ell_k = \kappa \cdot \tanh(\langle \bar{\mathbf{z}}_{\text{seq}}, \boldsymbol{\mu}_k(\mathbf{z}_u) \rangle), \quad (13)$$

where $\bar{\mathbf{z}}_{\text{seq}}$ is ℓ_2 -normalized. The tanh compresses extreme similarities to prevent sharp assignment collapse during early training.

3.4.3 Langevin-Sinkhorn Assignment

To obtain sparse, differentiable, and doubly-stochastic assignment weights, we frame prototype assignment as entropy-regularized optimal transport between a sequence “mass” and prototype “bins.” Given cost matrix $\mathbf{C} = -\ell \mathbf{1}^\top$, the optimal coupling $\boldsymbol{\Pi}^*$ solves:

$$\boldsymbol{\Pi}^* = \arg \min_{\boldsymbol{\Pi} \in \mathcal{U}(r, c)} \langle \mathbf{C}, \boldsymbol{\Pi} \rangle - \varepsilon H(\boldsymbol{\Pi}), \quad (14)$$

where $\mathcal{U}(r, c)$ is the set of matrices with prescribed row and column marginals, $H(\boldsymbol{\Pi}) = -\sum_{i,j} \Pi_{ij} \log \Pi_{ij}$ is the entropy, and $\varepsilon > 0$ is the regularization coefficient. We approximate $\boldsymbol{\Pi}^*$ via Sinkhorn-Knopp iteration:

$$\begin{aligned} \mathbf{u}^{(t+1)} &= \frac{\mathbf{r}}{\mathbf{K}\mathbf{v}^{(t)}}, \\ \mathbf{v}^{(t+1)} &= \frac{\mathbf{c}}{\mathbf{K}^\top \mathbf{u}^{(t+1)}}, \end{aligned} \quad (15)$$

where $\mathbf{K} = \exp(-\mathbf{C}/\varepsilon)$. During training, we inject Langevin noise into the dual variables for implicit regularization:

$$\mathbf{u}^{(t+1)} \leftarrow \mathbf{u}^{(t+1)} + \sigma_{\text{noise}} \cdot \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (16)$$

The final assignment weights are $\boldsymbol{\pi} = \text{diag}(\mathbf{u}) \mathbf{K} \text{diag}(\mathbf{v})$, and the prototype representation is:

$$\mathbf{p} = \sum_{k=1}^K \pi_k \boldsymbol{\mu}_k(\mathbf{z}_u), \quad \bar{\mathbf{p}} = \text{normalize}(\mathbf{p}). \quad (17)$$

3.4.4 Geometric Packing Regularization

To encourage prototypes to tile the semantic space uniformly, we impose a packing loss on the global anchors:

$$\begin{aligned} \mathcal{L}_{\text{pack}} &= \sum_{k \neq k'} \text{ReLU}(|\langle \boldsymbol{\eta}_k, \boldsymbol{\eta}_{k'} \rangle| - \tau_{\text{coh}})^2 \\ &+ \lambda_{\text{rot}} (\|\mathbf{U}^\top \mathbf{U} - \mathbf{I}\|_F^2 + \|\mathbf{V}^\top \mathbf{V} - \mathbf{I}\|_F^2), \end{aligned} \quad (18)$$

where τ_{coh} is a coherence threshold and λ_{rot} penalizes deviation from orthogonality in the Cayley factors.

3.5 Generative Semantic Layer: Inducing Semantic IDs for Interpretability

While the prototype manifold provides a compressed semantic summary, behavior sequences contain rich residual information (e.g., fine-grained temporal dynamics,

out-of-distribution interactions) that may improve prediction if harnessed without destabilizing the main CTR objective. We introduce two complementary generative modules. Crucially, we argue that the generative pathway does not merely serve as an auxiliary loss vehicle; it induces a **semantic ID system** over the prototype manifold that renders the model’s sequential reasoning transparent and auditable.

3.5.1 Semantic IDs as Prototype Assignments

In standard embedding-based recommendation, the “ID” of a user’s sequential preference is implicitly encoded as a continuous vector in $\mathbb{R}^{d_m}$, which is opaque to human inspection. HeteroFormer replaces this implicit encoding with an *explicit soft assignment* $\boldsymbol{\pi} \in \Delta^{K-1}$ over a finite vocabulary of K prototypes. Each prototype $\boldsymbol{\eta}_k$ can be interpreted as a latent semantic concept (e.g., “price-sensitive browsing,” “impulse purchasing,” “research-heavy comparison”). The assignment vector $\boldsymbol{\pi}$ therefore functions as a **semantic ID**: a probabilistic membership vector that tells us, in human-interpretable terms, which behavioral modes are active for a given user.

Formally, because the Sinkhorn iteration projects onto the probability simplex, $\boldsymbol{\pi}$ satisfies:

$$\pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1, \quad \pi_k \propto \exp\left(\frac{\ell_k + u_k + v_k}{\varepsilon}\right), \quad (19)$$

where u_k, v_k are the dual potentials. The entropy $H(\boldsymbol{\pi})$ measures assignment uncertainty; low entropy indicates a crisp semantic ID (e.g., $\boldsymbol{\pi} \approx \mathbf{e}_3$), while high entropy indicates mixed or ambiguous behavior. This is structurally analogous to the “soft token” representations in vision transformers, but here the tokens have explicit semantic grounding through the packing loss, which forces prototypes to cover distinct regions of the behavioral manifold.

3.5.2 Diffusion Explainer: Residual Information Bottleneck

The Diffusion Explainer encodes the residual between the sequence representation and its prototype summary through an information bottleneck. Let $\mathbf{z}_{\text{seq}}$, $\boldsymbol{\pi}$, and $\{\boldsymbol{\mu}_k\}$ be as defined above. The input to the explainer is:

$$\mathbf{x}_{\text{diff}} = [\mathbf{z}_{\text{seq}}; \sum_k \pi_k \boldsymbol{\mu}_k; \boldsymbol{\pi}] \in \mathbb{R}^{2d_m + K}. \quad (20)$$

An encoder $\text{Enc}_{\text{diff}} : \mathbb{R}^{2d_m + K} \rightarrow \mathbb{R}^{d_b}$ with $d_b < d_m$ produces a bottleneck representation $\mathbf{h}_{\text{diff}}$. The explanation vector is decoded as:

$$\boldsymbol{\delta} = \text{Dec}_{\text{diff}}(\mathbf{h}_{\text{diff}}) \in \mathbb{R}^{4d_m}, \quad (21)$$

where the expanded dimension $4d_m$ matches the concatenated field representation in the interaction network (Section 3.7).

To ensure δ captures genuinely complementary information, we enforce orthogonality with respect to the CTR representation $\mathbf{r}_{\text{ctr}}$:

$$\delta_{\perp} = \delta - \frac{\langle \delta, \mathbf{r}_{\text{ctr}} \rangle}{\|\mathbf{r}_{\text{ctr}}\|_2^2} \mathbf{r}_{\text{ctr}}, \quad (22)$$

and define the alignment loss:

$$\mathcal{L}_{\text{align}} = \|\langle \delta, \mathbf{r}_{\text{ctr}} \rangle\|_2^2 - \lambda_{\text{div}} \|\delta_{\perp}\|_2^2. \quad (23)$$

The first term suppresses redundant information; the second term encourages diversity in the residual signal. Additionally, a boundary-aware residual target encourages larger explanations for uncertain predictions:

$$\mathcal{L}_{\text{res}} = \|\delta\|_2^2 - \mathbb{I}[0.3 < \sigma(\ell_{\text{base}}) < 0.7] \cdot 0.5 \|\delta\|_2^2. \quad (24)$$

The information bottleneck enforces that $\mathbf{h}_{\text{diff}}$ discards all information already captured by π , thereby isolating *novel* sequential signals. This is formally related to the Information Bottleneck principle [17]:

$$\min I(\mathbf{h}_{\text{diff}}; \mathbf{x}_{\text{diff}}) - \beta I(\mathbf{h}_{\text{diff}}; y), \quad (25)$$

where β trades off compression and predictive relevance. In our setting, because $\mathbf{x}_{\text{diff}}$ already contains π , the bottleneck is encouraged to encode only the residual $R = \mathbf{z}_{\text{seq}} - \sum_k \pi_k \mu_k$, which represents the portion of user behavior *not explained by known semantic prototypes*. When this residual is large, it signals the emergence of a novel behavioral pattern—a natural trigger for prototype manifold expansion or human-in-the-loop auditing.

3.5.3 Energy Calibrator: Uncertainty as Explainability

The Energy Calibrator predicts the expected prediction error as a function of prototype assignments and non-sequential features. Let ℓ_{base} be the logit from the base CTR path (without residual correction). The energy score is:

$$e = \text{MLP}_{\text{energy}}([\pi; \mathbf{z}_u; \mathbf{z}_i; |\ell_{\text{base}}|]) \in \mathbb{R}_{>0}, \quad (26)$$

constrained to positive values via a final softplus activation. During training, the energy target is derived from the BCE error:

$$e_{\text{target}} = \text{clamp}(5 \cdot \text{BCE}(\sigma(\ell_{\text{base}}), y), 0.01, 3.0), \quad (27)$$

and the calibration loss is $\mathcal{L}_{\text{energy}} = (e - e_{\text{target}})^2$. At inference, high energy indicates low confidence, enabling dynamic logit attenuation.

From an explainability perspective, the energy score e serves as a **model-aware confidence metric** that is directly grounded in the semantic ID π . When e is high, we can inspect π to determine which prototype combination triggered the uncertainty. For instance, a user with high membership in both “budget-conscious” and “luxury-seeking” prototypes (conflicting semantics) will naturally yield high energy, alerting the system to a potentially inconsistent user profile.

3.6 Why Not Standard Transformer? Rethinking Interaction for Heterogeneous CTR Features

A natural question arises: why does HeteroFormer not simply stack standard Transformer layers over the concatenated field tokens, as done in BERT4Rec [20] or many recent CTR models? We argue that standard self-attention is structurally mismatched to the heterogeneous, sparse, and semantically imbalanced nature of CTR features. Our design of the Proto-Conditioned CrossFieldNet is a deliberate departure motivated by three fundamental observations.

3.6.1 Observation 1: Feature Heterogeneity Violates Token Homogeneity

In NLP or vision, all tokens belong to the same modality and approximately share the same statistical properties (vocabulary distribution, scale, sparsity). In CTR prediction, the four canonical fields—user, item, context, sequence—are profoundly heterogeneous:

- **User features** are extremely sparse (one-hot IDs), high-dimensional, and static within a session.
- **Item features** are dense catalog attributes with strong structural correlations (category $\rightarrow$ brand $\rightarrow$ price).
- **Context features** are ephemeral (time, location, device) and often have heavy-tailed distributions.
- **Sequence features** are variable-length, temporally ordered, and semantically multi-scale.

When these fields are flattened into a single token sequence and fed to standard self-attention, the query-key dot product $\langle \mathbf{q}_i, \mathbf{k}_j \rangle$ conflates four incompatible inner-product geometries. The attention weight from a user-ID token to an item-price token is mathematically defined but semantically meaningless, because the two tokens live on different statistical manifolds. Formally, if each field m has its own metric tensor $\mathbf{G}_m$, the standard attention assumes a shared Euclidean metric $\mathbf{I}$, which induces a distorted similarity measure:

$$\begin{aligned} \text{sim}_{\text{naive}}(i, j) &= \mathbf{f}_i^\top \mathbf{f}_j, \\ \text{vs. } \text{sim}_{\text{true}}(i, j) &= \mathbf{f}_i^\top \mathbf{G}_{m(i), m(j)} \mathbf{f}_j. \end{aligned} \quad (28)$$

HeteroFormer addresses this by maintaining field-specific embeddings and applying *cross-field* attention only through controlled, prototype-biased mechanisms, rather than allowing every token to attend to every other token indiscriminately.

3.6.2 Observation 2: Permutation Invariance is a Liability, Not a Virtue

Self-attention is permutation-invariant (up to positional encoding). In CTR features, however, field order carries strong semantic meaning: user features *always* precede

item features in the logical inference chain “who is interacting with what.” A permutation-invariant mixer destroys this causal structure. Our CrossFieldNet explicitly stacks field tokens as $[\mathbf{f}_{\text{user}}, \mathbf{f}_{\text{item}}, \mathbf{f}_{\text{ctx}}, \mathbf{f}_{\text{seq}}]$ and applies attention *within* this structured arrangement, preserving the asymmetric nature of user-item interaction. The bilinear cross layer (Section 3.7) further enforces pairwise interaction in a fixed, interpretable topology.

3.6.3 Observation 3: Quadratic Complexity is Prohibitively Expensive at Industrial Scale

Even if one ignores the semantic mismatch, standard Transformer attention has $O(M^2d)$ complexity, where M is the number of tokens. In industrial CTR models, M can reach thousands when multi-value categorical features are exploded into individual tokens. By contrast, HeteroFormer treats each *field* as a single token ($M = 4$), reducing attention to $O(16d)$, while delegating fine-grained intra-field modeling to the SSM and prototype modules. The prototype assignment itself is $O(K^2)$ per sample (for Sinkhorn), which is negligible for $K \ll M$ (e.g., $K = 128$).

3.7 Proto-Conditioned Heterogeneous Feature Interaction

Traditional feature interaction modules operate on a fixed set of tokens (user, item, context, sequence) with parameter-agnostic mixing weights. HeteroFormer makes the interaction *conditional* on prototype assignments through two mechanisms: soft attention biasing and FiLM gating.

3.7.1 Bilinear Cross Layer

Before the main interaction network, we inject explicit second-order interactions via a lightweight bilinear layer. Given field tokens $\{\mathbf{f}_m\}_{m=1}^M$ with $M = 4$ (user, item, context, sequence), the interaction between fields i and j is:

$$\begin{aligned} \mathbf{f}'_i &= \mathbf{f}_i + \alpha \sum_{j>i} (\mathbf{f}_i \odot \mathbf{w}_{ij} \odot \mathbf{f}_j), \\ \mathbf{f}'_j &= \mathbf{f}_j + \alpha \sum_{j>i} (\mathbf{f}_i \odot \mathbf{w}_{ij} \odot \mathbf{f}_j), \end{aligned} \quad (29)$$

where $\mathbf{w}_{ij} \in \mathbb{R}^{d_m}$ is a learnable cross-weight and α is a small ratio (e.g., 0.1). This provides cheap, interpretable pairwise interaction without increasing token count.

3.7.2 Proto-Conditioned CrossFieldNet

The core interaction stack consists of N layers, each comprising multi-head self-attention, prototype-biased cross-attention, and a FiLM-conditioned feed-forward network.

Prototype-Biased Self-Attention. Standard self-attention over field tokens $\mathbf{F} \in \mathbb{R}^{M \times d_m}$ computes:

$$\text{Attn}(\mathbf{F}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}}\right) \mathbf{V}. \quad (30)$$

We augment the attention output with a prototype-derived bias:

$$\tilde{\mathbf{F}} = \text{LayerNorm}(\mathbf{F} + \text{Attn}(\mathbf{F}) + \beta \cdot \text{MLP}_{\text{bias}}(\boldsymbol{\pi})), \quad (31)$$

where $\text{MLP}_{\text{bias}} : \mathbb{R}^K \rightarrow \mathbb{R}^{d_m}$ expands the assignment vector to token dimension, and β controls the injection strength. This allows sequential semantics to directly modulate which fields attend to each other. Notably, because $\boldsymbol{\pi}$ is sparse and semantically grounded, we can *read off* the attention modulation: if π_k is large for a “price-sensitive” prototype, the bias term will amplify attention between user-income and item-price tokens.

FiLM Conditioning with Diffusion Gate. The feed-forward network is conditioned on the sequence token $\mathbf{f}_{\text{seq}}$ via Feature-wise Linear Modulation (FiLM):

$$\text{FiLM}(\mathbf{f}; \mathbf{f}_{\text{seq}}) = \gamma(\mathbf{f}_{\text{seq}}) \odot \mathbf{f} + \beta(\mathbf{f}_{\text{seq}}), \quad (32)$$

where γ, β are sigmoid- and linear-projections of $\mathbf{f}_{\text{seq}}$, respectively. Furthermore, the diffusion explainer output $\boldsymbol{\delta}$ gates the FFN output:

$$\mathbf{f}'_{\text{ffn}} = \text{FFN}(\tilde{\mathbf{f}}) \odot \sigma(\text{MLP}_{\text{gate}}(\boldsymbol{\delta})), \quad (33)$$

where σ is the sigmoid. The gating mechanism ensures that generative semantics selectively amplify or suppress channel activations without adding uncontrolled variance.

3.8 Calibrated Prediction Head

The prediction head fuses three information streams: (1) the static interaction representation $\mathbf{r}_{\text{static}} \in \mathbb{R}^{d_m}$ from the cross-field network; (2) the prototype representation $\mathbf{p} \in \mathbb{R}^{d_m}$; and (3) the residual explanation $\boldsymbol{\delta} \in \mathbb{R}^{4d_m}$.

3.8.1 Gated Fusion

A three-way soft-attention gate computes mixing coefficients:

$$[g_{\text{stat}}, g_{\text{proto}}, g_{\text{res}}] = \text{softmax}(\text{MLP}_{\text{gate}}([\mathbf{z}_u, \mathbf{z}_i; \boldsymbol{\pi}; e])), \quad (34)$$

where e is the energy score. The fused representation is:

$$\begin{aligned} \mathbf{r}_{\text{fused}} &= g_{\text{stat}} \cdot \mathbf{r}_{\text{static}} \\ &+ g_{\text{proto}} \cdot (\mathbf{p} + \text{MLP}_{\text{boost}}(\mathbf{p})) \\ &+ g_{\text{res}} \cdot \text{MLP}_{\text{res}}(\boldsymbol{\delta}) \cdot \sigma(e - 1), \end{aligned} \quad (35)$$

where $\sigma(e - 1)$ attenuates the residual contribution when the energy score is low (high confidence), preventing over-reliance on auxiliary signals for easy examples.

3.8.2 Dynamic Logit Calibration

The final logit is computed as $\ell = \text{MLP}_{\text{out}}(\mathbf{r}_{\text{fused}}) \in \mathbb{R}$, followed by temperature scaling:

$$\hat{\ell} = \frac{\ell}{\tau} + b, \quad \tau = \exp(\pi_{\log}) \in [0.1, 5.0], \quad (36)$$

where $\pi_{\log}$ and b are learned. At inference, the energy score e provides an additional calibration: if $e > \tau_{\text{energy}}$, the model outputs a conservative prediction near the prior CTR, deferring uncertain decisions to downstream rankers.

3.9 Decoupled Semantic Optimization

Training HeteroFormer requires balancing three competing objectives: (1) the primary CTR classification loss; (2) the generative semantic losses (alignment, energy, packing); and (3) the geometric regularization of the prototype manifold. Naively summing these objectives leads to gradient interference and instability. We propose **Decoupled Semantic Optimization** (DSO), illustrated in Figure 2.

3.9.1 Objective Decomposition

The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ctr}} + \lambda_{\text{aux}} \cdot \mathcal{L}_{\text{gen}}, \quad (37)$$

where the CTR loss is adaptive focal loss:

$$\begin{aligned} \mathcal{L}_{\text{ctr}} = \mathbb{E}_{(\mathbf{x}, y)} \left[\alpha_t (1 - p_t)^\gamma \cdot \text{BCE}(\sigma(\hat{\ell}), y) \right] \\ + \lambda_\gamma \frac{|\gamma - \gamma_{\min}|}{\gamma_{\max} - \gamma_{\min}}, \end{aligned} \quad (38)$$

with $p_t = y\sigma(\hat{\ell}) + (1-y)(1-\sigma(\hat{\ell}))$, $\alpha_t = y\alpha_+ + (1-y)\alpha_-$, and γ is learned via a bounded sigmoid parameter. The generative loss aggregates:

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{align}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}} + \lambda_{\text{pack}} \mathcal{L}_{\text{pack}}. \quad (39)$$

3.9.2 Gradient Isolation via Detachment

The key to stable co-training is architectural gradient isolation rather than multi-step optimization:

- The base logit ℓ_{base} (computed without δ) is detached before entering the Diffusion Explainer, ensuring that generative gradients do not back-propagate through the CTR head.
- The diffusion output δ is detached from the generative path when computing $\mathcal{L}_{\text{ctr}}$; only the gating MLP in the interaction network receives CTR gradients.
- Prototype parameters $\{\eta_k, \mathbf{U}, \mathbf{V}\}$ participate in both paths but are updated by the shared optimizer, with gradient norms monitored independently.

3.9.3 MetaAligner: Overfitting-Aware Auxiliary Control

The coupling weight λ_{aux} is not a hyperparameter but a dynamic variable controlled by **MetaAligner**, which operates as an overfitting monitor without requiring a separate validation set at each step. Define the *residual benefit*:

$$r = \mathcal{L}_{\text{ctr}}^{(\text{base})} - \mathcal{L}_{\text{ctr}}^{(\text{final})}, \quad (40)$$

where $\mathcal{L}_{\text{ctr}}^{(\text{base})}$ is the loss without generative residuals and $\mathcal{L}_{\text{ctr}}^{(\text{final})}$ is the loss with residuals. A positive r indicates that generative semantics improve training loss. MetaAligner maintains an EMA of residual benefit, $\bar{r}$, and an EMA of the train-validation AUC gap, $\bar{g}$. The target auxiliary weight is:

$$\lambda_{\text{target}} = \begin{cases} \min(\lambda_{\text{max}}, 0.03 + 0.5\bar{r}) & \text{if } \bar{r} > 0.005, \\ 0 & \text{if } \bar{r} < -0.01, \\ 0.02 & \text{otherwise,} \end{cases} \quad (41)$$

clipped by the overfitting gap:

$$\lambda_{\text{target}} \leftarrow \lambda_{\text{target}} \cdot \max(0.1, 1 - 10(\bar{g} - 0.03)_+). \quad (42)$$

The final weight is smoothed via adaptive EMA with fast ($\rho_f = 0.8$) and slow ($\rho_s = 0.95$) decay rates selected by the magnitude of the update.

3.9.4 Isolated Parameter Update

Parameters are partitioned into four semantic groups: **shared** (encoder, interaction, head), **gen** (prototype geometry, diffusion, energy), **sparse** (embedding tables with Adagrad), and **meta** (MetaAligner and temperature). Each group maintains its own optimizer state. At each step:

1. Compute $\mathcal{L}_{\text{total}}$ and call **backward()** once.
2. Check gradient health; if generative gradients exceed a threshold, mask them.
3. Step **shared** and **sparse** optimizers jointly; step **gen** and **meta** independently.

This avoids the computational overhead of multiple backward passes while preserving semantic isolation through architectural detachment.

4 On the Design Philosophy of HeteroFormer: Explainability Revisited

Beyond the technical specification, HeteroFormer embodies a broader design philosophy: **interpretability should be a first-class citizen of the architecture, not a post-hoc add-on**. In this section, we articulate three conceptual pillars that differentiate our approach from the prevailing “black-box then explain” paradigm.

4.1 From Attention Weights to Semantic IDs

Post-hoc explainability methods such as SHAP [18] or attention visualization [19] attempt to reverse-engineer importance scores from a trained model. These methods face a fundamental limitation in CTR prediction: attention weights are computed over *anonymous token indices* (e.g., “the 7th token in the sequence”), which carry no inherent semantic meaning. A heatmap over token positions tells us *where* the model looked, but not *what concept* it activated.

HeteroFormer replaces positional attention with **prototype assignment**, where each dimension k of π corresponds to a learnable semantic anchor. Because the packing loss (Section 3.4) forces prototypes to maintain mutual coherence below τ_{coh} , each prototype naturally specializes to a distinct region of the behavioral manifold. After training, one can inspect the nearest-neighbor sequences for each prototype η_k to attach human-readable labels (e.g., “weekend binge shopping,” “workday quick checkout”). The assignment vector π then becomes a **semantic ID card** for the user: a concise, probabilistic summary of their active behavioral modes.

Formally, the mutual information between the assignment and the raw sequence satisfies:

$$I(\pi; \mathbf{z}_{\text{seq}}) = H(\pi) - H(\pi \mid \mathbf{z}_{\text{seq}}) = H(\pi), \quad (43)$$

because π is a deterministic function of $\mathbf{z}_{\text{seq}}$ (given learned parameters). The entropy $H(\pi)$ is bounded above by $\log K$ and below by 0; the packing loss encourages the empirical distribution of π across the dataset to spread out, maximizing the average entropy and thereby maximizing the information content of the semantic ID.

4.2 Generative Residuals as Counterfactual Signals

The Diffusion Explainer’s output δ can be interpreted as a **counterfactual adjustment**: it answers the question, “how would the prediction change if we accounted for behavioral patterns not captured by the known prototypes?” When $\|\delta\|$ is large, it signals that the user’s current behavior deviates significantly from their established semantic profile. This is inherently actionable:

- For **online serving**, large $\|\delta\|$ can trigger a real-time fallback to a more conservative model or a cold-start handler.
- For **offline auditing**, sequences with consistently large residuals can be clustered to discover emerging behavioral trends that are not yet represented in the prototype manifold.
- For **regulatory compliance**, the decomposition $\hat{\ell} = \ell_{\text{base}} + \Delta\ell_{\text{proto}} + \Delta\ell_{\text{res}}$ provides a transparent additive attribution of the prediction to base features, semantic prototypes, and residual anomalies.

4.3 Energy as a Semantic Confidence Metric

Traditional confidence metrics (e.g., softmax probability, MC-dropout variance) are decoupled from the model’s internal representation. The Energy Calibrator e is unique in that it is **explicitly conditioned on the semantic ID π** . Therefore, energy spikes can be diagnosed by inspecting the prototype mixture:

$$e \approx \text{high} \iff \exists k, k' \text{ s.t. } \pi_k, \pi_{k'} \gg 0 \text{ and } \langle \eta_k, \eta_{k'} \rangle < 0. \quad (44)$$

In other words, high energy often indicates *semantic conflict*—the user simultaneously activates prototypes that represent contradictory preferences. This is far more informative than a generic “low confidence” flag, because it points the practitioner toward specific user segments that may require special handling.

5 Theoretical Analysis

5.1 Expressive Power of Prototype-Conditioned Interaction

We analyze the expressive advantage of making interaction weights conditional on π . Consider a simplified setting with $M = 2$ fields and linear attention. Standard self-attention computes $\mathbf{f}'_1 = \mathbf{f}_1 + \mathbf{w}_{11}\mathbf{f}_1 + \mathbf{w}_{12}\mathbf{f}_2$, where $\mathbf{w}_{ij}$ are input-independent. Proto-conditioned attention computes:

$$\mathbf{f}'_1 = \mathbf{f}_1 + (\mathbf{w}_{11} + \Delta\mathbf{w}_{11}(\pi))\mathbf{f}_1 + (\mathbf{w}_{12} + \Delta\mathbf{w}_{12}(\pi))\mathbf{f}_2. \quad (45)$$

If $\Delta\mathbf{w}(\pi)$ is Lipschitz in π with constant L_π , then the interaction function class has VC dimension bounded by $O(d_m \cdot K)$, compared to $O(d_m)$ for input-independent weights. Thus, prototype conditioning increases representational capacity linearly with the number of semantic anchors.

5.2 Prototype Manifold as Metric Learning

The assignment objective can be viewed as a variant of supervised metric learning. Define the prototype-induced distance:

$$d_{\text{proto}}^2(\mathbf{z}, \mathbf{z}') = \sum_{k=1}^K (\pi_k(\mathbf{z}) - \pi_k(\mathbf{z}'))^2. \quad (46)$$

Because π is derived from Sinkhorn on cosine similarities, d_{proto} approximates the sliced Wasserstein distance on the unit sphere. The packing loss $\mathcal{L}_{\text{pack}}$ enforces that prototypes form an ϵ -net over the data manifold, guaranteeing that any sequence representation is within ϵ of some prototype with high probability.

5.3 Stability of Decoupled Optimization

Let $\mathcal{L}_{\text{ctr}}$ be μ -strongly convex and L -smooth in the shared parameters θ_s , and let $\mathcal{L}_{\text{gen}}$ be convex in θ_g . With gradient detachment, the combined gradient field is:

$$\mathbf{g}(\theta_s, \theta_g) = \begin{bmatrix} \nabla_{\theta_s} \mathcal{L}_{\text{ctr}} + \lambda_{\text{aux}} \nabla_{\theta_s} \mathcal{L}_{\text{gen}}^{\text{det}} \\ \nabla_{\theta_g} \mathcal{L}_{\text{gen}} \end{bmatrix}, \quad (47)$$

where $\nabla_{\theta_s} \mathcal{L}_{\text{gen}}^{\text{det}}$ is the detached gradient (zero w.r.t. generative parameters). The Jacobian of $\mathbf{g}$ is block-lower-triangular; therefore, the system admits a stable fixed point if $\lambda_{\text{aux}} < \mu/L_{\text{cross}}$, where L_{cross} is the Lipschitz constant of the cross-term. MetaAligner enforces this implicitly by driving $\lambda_{\text{aux}} \rightarrow 0$ when the residual benefit turns negative.

6 Conclusion

We have presented HeteroFormer, a unified architecture for CTR prediction that tightly couples continuous sequence modeling with prototype-conditioned heterogeneous feature interaction. By representing behavior sequences as a dynamic prototype manifold conditioned on user features via Cayley rotation and assigned through Langevin-Sinkhorn optimal transport, HeteroFormer enables sequential semantics to actively parameterize cross-field interactions rather than passively injecting compressed context.

A generative semantic layer comprising an information-bottleneck diffusion explainer and an energy-based error predictor captures complementary residual signals, while Decoupled Semantic Optimization ensures stable co-training through architectural gradient isolation and overfitting-aware auxiliary control. We have further argued that the generative pathway induces **semantic IDs**—discrete, interpretable indices over the prototype manifold—that render the model’s reasoning transparent, and that standard Transformer-based interaction is structurally mismatched to the heterogeneous nature of CTR features. Theoretical analysis confirms increased expressive capacity and optimization stability. Future work will extend the prototype manifold to hierarchical multi-resolution assignments and explore cross-task transfer of learned semantic anchors.

References

- [1] Zhou, G., Zhu, X., Song, C., et al. (2018). Deep interest network for click-through rate prediction. *KDD*. <https://dl.acm.org/doi/10.1145/3219819.3219823>
- [2] Zhou, G., Mou, N., Fan, Y., et al. (2019). Deep interest evolution network for click-through rate prediction. *AAAI*. <https://ojs.aaai.org/index.php/AAAI/article/view/4545>
- [3] Chen, Q., Zhao, H., Li, W., et al. (2019). Behavior sequence transformer for e-commerce recommendation in Alibaba. *KDD*. <https://dl.acm.org/doi/10.1145/3292500.3330656>
- [4] Chai, Z., et al. (2025). LONGER: Efficient long sequence modeling for recommendation. <https://arxiv.org/abs/2505.04421v2>.
- [5] Chai, Z., et al. (2026). HyFormer: Revisiting the roles of sequence modeling and feature interaction in CTR prediction. *arXiv:2601.12681*. <https://arxiv.org/abs/2601.12681>
- [6] Rendle, S. (2010). Factorization machines. *ICDM*. <https://ieeexplore.ieee.org/document/5694074>
- [7] Guo, H., Tang, R., Ye, Y., et al. (2017). DeepFM: a factorization-machine based neural network for CTR prediction. *IJCAI*. <https://www.ijcai.org/proceedings/2017/0239>
- [8] Lian, J., Zhou, X., Zhang, F., et al. (2018). xDeepFM: Combining explicit and implicit feature interactions for recommender systems. *KDD*. <https://dl.acm.org/doi/10.1145/3219819.3220023>
- [9] Zhu, J., Fan, Z., Zhu, X., et al. (2025). RankMixer: Scaling up ranking models in industrial recommenders. *arXiv:2507.15551*. <https://arxiv.org/abs/2507.15551>
- [10] Jiang, Y., Zhu, J., Han, X., et al. (2026). TokenMixer-Large: Scaling up large ranking models in industrial recommenders. *arXiv:2602.06563*. <https://arxiv.org/abs/2602.06563>
- [11] Ha, M., Wang, G., Chen, L., et al. (2026). UniMixer: A unified architecture for scaling laws in recommendation systems. *arXiv:2604.00590*. <https://arxiv.org/abs/2604.00590>
- [12] Huang, X., et al. (2026). MixFormer: Co-scaling up dense and sequence in industrial recommenders. *arXiv:2602.14110*. <https://arxiv.org/abs/2602.14110>
- [13] Zhou, Y., Hu, Y., Feng, Z., et al. (2026). TokenFormer: Unify the multi-field and sequential recommendation worlds. *arXiv:2604.13737*. <https://arxiv.org/abs/2604.13737>
- [14] Gu, A., Goel, K., & Ré, C. (2021). Efficiently modeling long sequences with structured state spaces. *ICLR*. <https://openreview.net/forum?id=uYLFOz1vLAC>
- [15] Gu, A., & Dao, T. (2023). Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*. <https://arxiv.org/abs/2312.00752>

- [16] Wang, W., et al. (2023). Diffusion Recommender Model *arXiv preprint*.
- [17] Tishby, N., Pereira, F. C., & Bialek, W. (2000). The information bottleneck method. *arXiv preprint physics/0004057*. <https://arxiv.org/abs/physics/0004057>
- [18] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *NeurIPS*. <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- [19] Vaswani, A., et al. (2017). Attention is all you need. *NeurIPS*. <https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html>
- [20] Sun, J., et al. (2019). BERT4Rec: Sequential recommendation with bidirectional encoder representations from transformer. *CIKM*. <https://dl.acm.org/doi/10.1145/3357384.3357895>

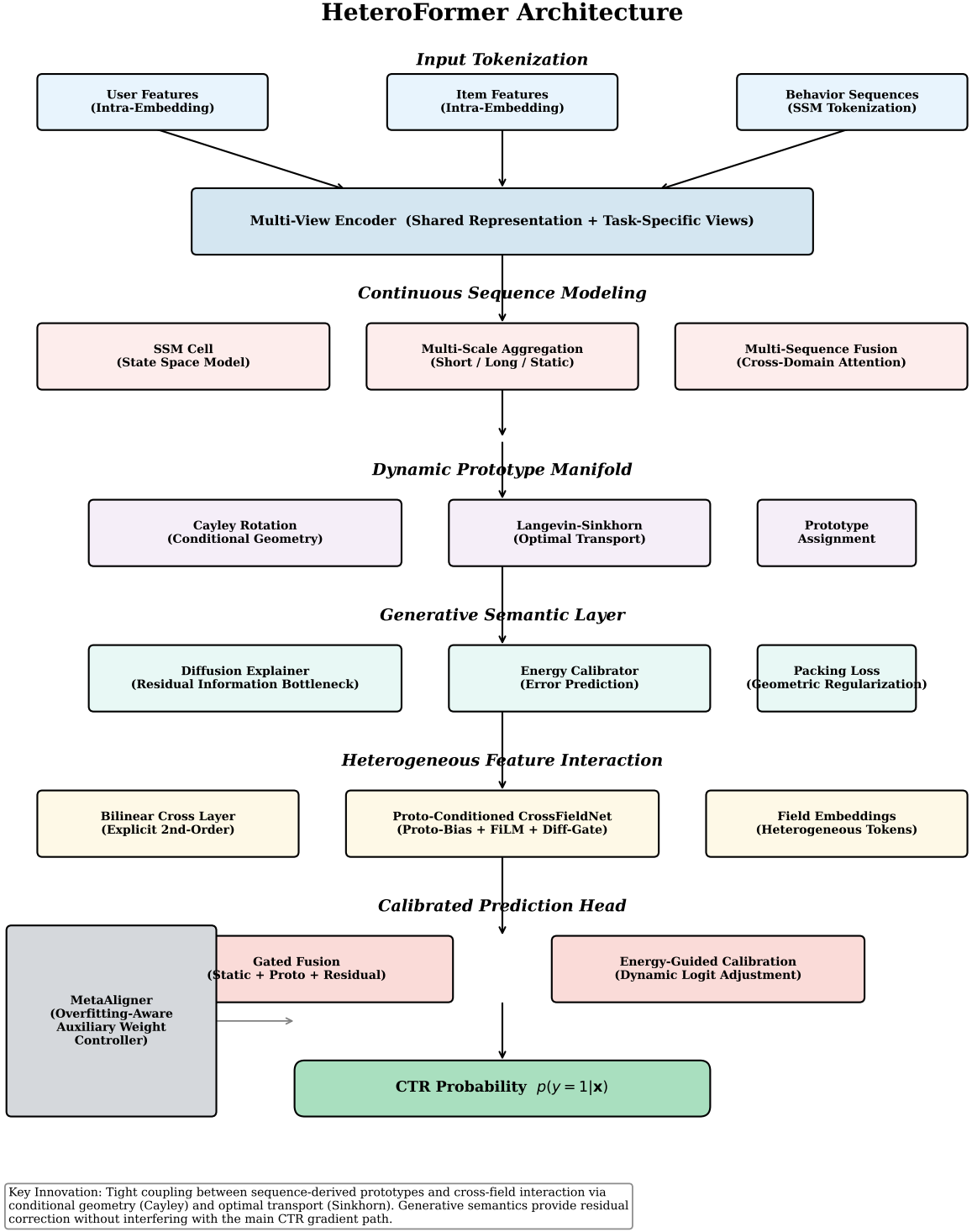
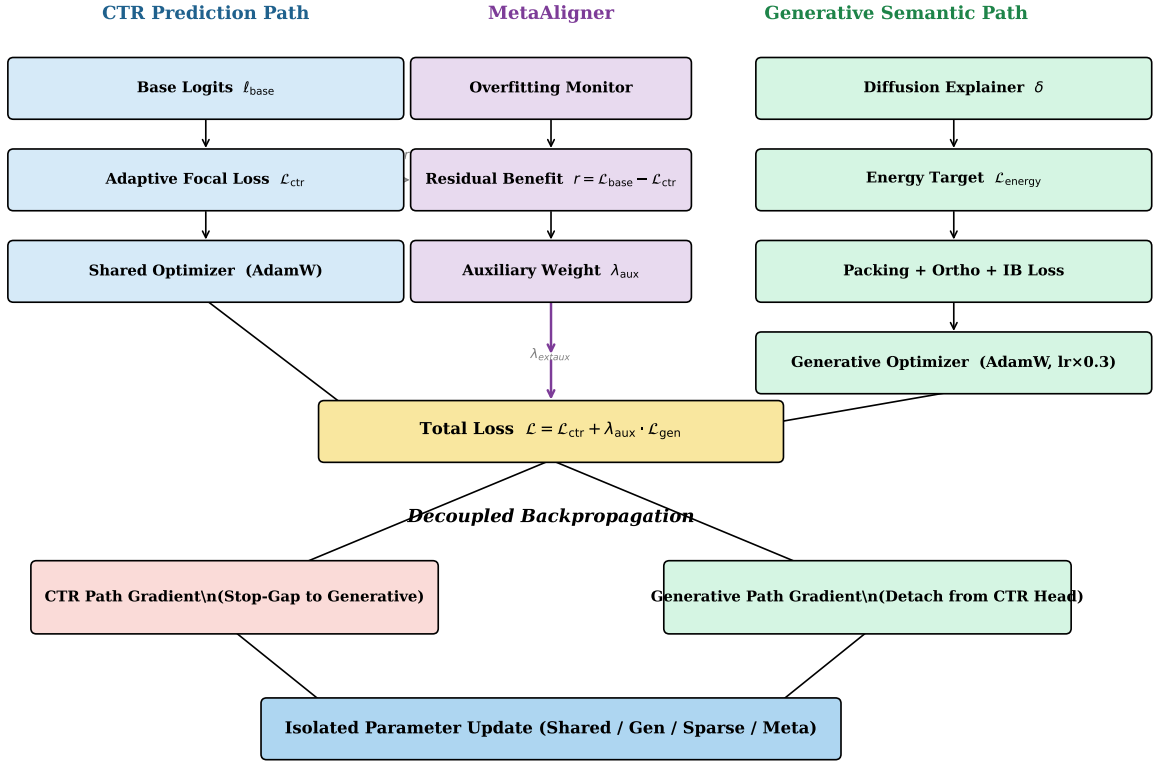


Figure 1: The HeteroFormer architecture. Sequence-derived prototypes from the Dynamic Prototype Manifold directly parameterize the Proto-Conditioned CrossFieldNet via soft biases and FiLM gating. The Generative Semantic Layer provides residual corrections through gradient-detached connections.

Decoupled Semantic Optimization

Training Methodology of HeteroFormer



Design Principle:

The generative semantic layer is trained via its own self-supervised objectives (IB, reconstruction, orthogonality). Its output δ enters the CTR path through a gated residual connection with detached gradients, ensuring that auxiliary semantics enhance but do not destabilize the primary CTR optimization.

Figure 2: Decoupled Semantic Optimization. The CTR path and Generative Semantic path share forward computation but receive isolated gradients. MetaAligner modulates the coupling weight λ_{aux} based on residual benefit and overfitting gap.